

The Biosciences and Bioengineering faculty at IIT Bombay are collectively pursuing an exceptionally broad and deeply specialized portfolio of research spanning molecular to systems-level biology, synthetic and systems biology, biophysics, biomedical engineering, and translational medicine. Their current and recent work includes:

- Mathematical and computational modeling of physiological processes and disease dynamics, such as deformable-lung ventilator waveform modeling, protease-based enzymatic biosensor design, and noninvasive ventilator-dyssynchrony detection;
- Development of signal-exchange architectures for synthetic biological control circuits, including modular protein-oligonucleotide communication systems and biomolecular integral controllers implementing robust gene regulation;
- Carbohydrate-structure elucidation and glycomics, with one-pot synthesis of bacterial capsular trisaccharides for vaccine development, mechanistic annotation of left-handed β -helix domains, and construction of comprehensive monosaccharide-biosynthesis pathway databases;
- Molecular mechanisms of bacterial pathogenesis and host-pathogen interactions, including differential ubiquitination strategies at the blood-brain barrier, pneumococcal transcytosis determinants, pore-forming toxin adaptation for intracellular survival, and membrane-vesicle-mediated disruption of maternal-fetal barriers;
- Structure-based enzymology targeting malaria and other infectious diseases, exemplified by zymogen inactivation pathways in Plasmepsin X, structure-function analyses of glutamate dehydrogenase inhibition, activation mechanisms of aspartic proteases, and structure-guided design of peptidomimetic inhibitors;
- Chromosome-segregation mechanics and epigenetic regulation in meiosis and mitosis, probing 14-3-3-mediated kinetochore integrity, SWI/SNF interactions in plasmid hitchhiking, kinesin-driven kinetochore assembly, and shugoshin's role in spindle-assembly checkpoint maintenance;
- Single-particle cryo-EM and cryo-ET investigations of ribosome-protein interactions and translation-termination dynamics via time-resolved structural snapshots;
- Advanced biomedical imaging and instrumentation, including full-view LED-based optoacoustic tomography, spiral volumetric tracking of nanoparticle kinetics, head-to-tail small-animal imaging, and multimodal deep-learning reconstructions for breast cancer and rheumatoid-arthritis screening;
- Motor-control neurosciences and rehabilitation science, investigating anticipatory and compensatory postural adjustments in aging, high-intensity interval training effects on post-stroke corticomotor excitability, and noninvasive brain-stimulation protocols for movement disorders;
- Molecular virology and environmental virology, elucidating replication, repair, and packaging in giant viruses, leveraging MMP modulation for embryonic stem-cell differentiation, and developing PCB-electrode-based electrochemical viral-RNA detection;
- Protein-dynamics studies via solution and solid-state NMR, mapping disorder-mediated chromatin organization, STD-methyl TROSY epitope mapping of peptide drugs, and characterizing protein-lipid and protein-drug interactions;
- Cancer-stem-cell biology and chemoresistance mechanisms in breast cancer, focusing on quiescent DLL1+ tumor cells, Th9-axis-mediated oxidative-stress modulation in T-cell

lymphoma, and multiparametric big-data analyses for biomarker discovery;

- Biophysical and computational studies of molecular motors and cytoskeletal dynamics, including Monte Carlo and MD simulations of kinesin and dynein transport, Nagaoka ferromagnetism in motor arrays, and mechanistic insights into motor-tail drift effects on collective transport;
- Amyloid formation in neurodegeneration and oncology, dissecting p53-amyloid-driven oncogenic gain-of-function, designing de novo functional amyloids for drug delivery, and studying co-aggregation processes in functional prolactin–alanine amyloids;
- Intracellular transport and lipid-droplet–ER contact regulation, reconstituting metabolic and immune-sensitive membrane contacts, mapping sphingolipid flux during phagosomal maturation, and elucidating dynein–dynactin control mechanisms on native cargoes;
- Computational neuroscience models of stretch-induced calcium signaling in umbrella cells and medium-spiny neurons, leveraging the NEURON platform to predict smooth-muscle electrophysiology and synaptic-transmission dynamics;
- Fiber-optic and plasmonic-LSPR biosensors for detection of antibiotics, heavy metals, pathogens, and biomarkers in clinical and environmental samples, integrating dendrimer-capped nanoparticles, enzymatic hydrolysis on nanostructured polyaniline, and label-free bacteriophage functionalization;
- Mechanistic studies of embryonic development using zebrafish models to define dynamic cell-size optima, ploidy-manipulation screens, and resilience windows minimizing teratogenic heat effects;
- Chromatin biophysics, employing polymer coarse-graining models to predict scale-dependent chromatin properties, information-theoretic models of histone-modification inheritance, and soft-matter statistical physics applied to nuclear architecture;
- Anti-microbial drug-target discovery against *Plasmodium falciparum* via tubulin-targeting compounds, antisense-oligonucleotide modulation of siRNA/miRNA activity, and proteomics-driven malaria biomarker identification;
- FtsZ dynamics and hyperthermia-induced cell-cycle arrests as antitumor mechanisms, including discovery of microtubule-targeting agents that impair kinesin-2–dependent β -catenin transport;
- Microfluidic diagnostics and therapeutics platforms, from centrifugal systems for large-volume processing to PNA-DiSc₂ absorbance assays for nucleic-acid detection, coupled with patent-registered paperfluidic devices for oral-health monitoring;
- Biodegradation and metabolic engineering of aromatic compounds in *Pseudomonas*, characterizing novel oxygenases and integrative conjugative elements facilitating adaptive colonization;
- Tumor-immunology engineering of CAR-T therapeutics' immunogenicity and Th9-axis survival pathways in cutaneous T-cell lymphoma;
- Mechanobiology of cell–ECM interactions, exploring geometric and topographical force cues governing cell sorting, proliferation, differentiation, and angiogenesis;
- Cilia and flagella assembly/disassembly and ubiquitin-mediated receptor trafficking for GPCR removal, including ESCRT-based cargo retrieval and SNX27-retromer regulation of GLUT recycling;
- Proteomic workflows for cancer and infectious-disease biomarker discovery, leveraging

2D-DIGE, mass spectrometry, and protein microarrays on longitudinal clinical samples;

- Novel optical-tomography inverse algorithms and speckle-contrast-based diffuse-correlation spectroscopy to enable low-cost, deep-tissue blood-flow measurements;
- Material-based in situ genetic engineering platforms using acellular scaffolds and lentiviral vectors for T-cell immunotherapy, macroporous cryogel matrices for off-the-shelf skin regeneration, and 3D spheroid models preserving tumor microenvironments for drug screening;
- Inverse molecular modeling of eccrine sweat composition, integrating molecular-dynamics simulations with experimental validation;
- Regulatory-omics of malaria and toxoplasmosis via importin- α mechanisms and nanoparticle carriers to improve antimalarial bioavailability;
- Advanced acoustic-microcalorimetric explosive sensing and noninvasive pesticide detection using conductive-polymer paper sensors;
- High-throughput SELEX and computational pipelines to down-regulate pathogenic miRNAs in human cell lines;
- Biomanufacturing intensification through continuous microfluidic platforms for nanoreactor-based synthesis of redox-active organics for flow-battery applications.

This exhaustive superset captures the full spectrum of highly specialized, current and recent research themes across the BSBE faculty at IIT Bombay, reflecting projects where students contribute from computational modeling to experimental design, instrumentation, and translational studies.

1. <https://ppl-ai-file-upload.s3.amazonaws.com/web/direct-files/attachments/126871353/32859864-4dd3-4e16-ae43-2a8145866cf1/bsbe.pdf>